

Theory and Methodology

Multiobjective programming in optimization of the interval objective function

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Abstract: In this paper, the mathematical programming problem whose objective function has interval coefficients is investigated. To maximize the interval objective function, the order relations which represent the decision maker's preference between interval profits are defined by the right limit, the left limit, the center and the width of an interval. The maximization problem with the interval objective function is converted into a multiobjective problem using the order relations. The minimization problem is also converted into a multiobjective problem using the order relations which represent the decision maker's preference between interval costs.

Keywords: Interval analysis, inexact programming, optimization, multiple criteria programming, decision theory

1. Introduction

In conventional mathematical programming, the coefficients of problems are assumed to be deterministic and fixed in value. But there are many situations where this assumption is not valid because of uncertain environments. Thus the decision making methods under uncertainty are needed. The stochastic programming and the fuzzy programming were proposed from this viewpoint. In stochastic programming problems [4,7], the coefficients are viewed as random variables and it is assumed that their probability distributions are known. On the other hand, in fuzzy programming problems [2,9] the constraints and the goals are viewed as fuzzy sets. It is also assumed that their

membership functions are known. Whereas the probability distributions and the membership functions play important roles in each method, it is not always easy for the decision maker to specify them.

In this paper, we intend to propose the simple decision making method under uncertainty by considering the coefficients in mathematical programming problems as intervals. The mathematical programming problem to be investigated is

$$\max(Z(\mathbf{x}) = A_1x_1 + A_2x_2 + \cdots + A_nx_n) \\ : \mathbf{x} \in S \subset R^n), \quad (1.1)$$

where S is a feasible region of $\mathbf{x}$ and A_i is an interval which represents the uncertain unit profit from x_i . This problem is one of the inexact programming problems [5,8]. If we take the conservative strategy by Falk [5], the maximization prob-

Received March 1989; revised April 1989

lem (1.1) may be reduced to the following problem:

$$\max \left(\left(\min_{a_i \in A_i} z(x) = a_1 x_1 + a_2 x_2 + \cdots + a_n x_n \right) : x \in S \subset R^n \right). \quad (1.2)$$

The solution of (1.2) is a max–min solution which is optimal against the worst case, that is, against the left limit of $Z(x)$ in (1.1).

In this paper we consider not only the left limit but also the right limit, the center and the width of the interval objective function $Z(x)$. **The innovative result of this paper is converting the maximization problem (1.1) into a multiobjective problem and defining the solution set of (1.1) by the Pareto optimal solutions of the multiobjective problem.** In order to reformulate (1.1) as the multiobjective problem, the order relations which represent the decision maker's preference between interval profits are defined by the left limit, the right limit, the center and the width of an interval.

The minimization problem with an interval objective function,

$$\min (Z(x) = A_1 x_1 + A_2 x_2 + \cdots + A_n x_n : x \in S \subset R^n), \quad (1.3)$$

is also investigated and converted into a multiobjective problem using the order relations which represent the decision maker's preference between interval costs.

This paper is organized as follows. In Section 2 notations of the interval and the interval arithmetics are briefly explained. In Section 3 the order relations between intervals are introduced for the maximization problem (1.1) and for the minimization problem (1.3) respectively. Some properties of those order relations are shown. In Section 4 the maximization problem (1.1) and the minimization problem (1.3) are reformulated as multiobjective problems using the order relations defined in Section 3 respectively. In Section 5 some numerical examples are demonstrated to explain the proposed method.

2. Interval arithmetic

Throughout this paper lower case letters denote real numbers and upper case letters denote closed

intervals. The set of all real numbers is denoted by R . An interval is defined by an ordered pair in brackets as

$$A = [a_L, a_R] = \{a : a_L \leq a \leq a_R, a \in R\}, \quad (2.1)$$

where a_L is the left limit and a_R is the right limit of A . The interval is also denoted by its center and width as

$$A = \langle a_C, a_W \rangle = \{a : a_C - a_W \leq a \leq a_C + a_W, a \in R\}, \quad (2.2)$$

where a_C is the center and a_W is the width of A . From (2.1) and (2.2), the center and width of an interval may be calculated as

$$a_C = \frac{1}{2}(a_R + a_L), \quad (2.3)$$

$$a_W = \frac{1}{2}(a_R - a_L). \quad (2.4)$$

The generalization of ordinary arithmetic to closed intervals is known as interval arithmetic. For complete details, see Alefeld and Herzberger [1], and Moore [6]. The basic definition [1] is as follows.

Definition 2.1. Let $*$ $\in \{+, -, \cdot, \div\}$ be a binary operation on the set of real numbers. If A and B are closed intervals, then

$$A * B = \{a * b : a \in A, b \in B\} \quad (2.5)$$

defines a binary operation on the set of closed intervals. In the case of division, it is assumed that $0 \notin B$.

The operations on intervals used in this paper may be explicitly calculated from Definition 2.1 as

$$\begin{aligned} A + B &= [a_L, a_R] + [b_L, b_R] \\ &= [a_L + b_L, a_R + b_R], \end{aligned} \quad (2.6)$$

$$\begin{aligned} A + B &= \langle a_C, a_W \rangle + \langle b_C, b_W \rangle \\ &= \langle a_C + b_C, a_W + b_W \rangle, \end{aligned} \quad (2.7)$$

$$kA = k[a_L, a_R] = \begin{cases} [ka_L, ka_R] & \text{for } k \geq 0, \\ [ka_R, ka_L] & \text{for } k < 0, \end{cases} \quad (2.8)$$

$$kA = k\langle a_C, a_W \rangle = \langle ka_C, |k| a_W \rangle, \quad (2.9)$$

where k is a real number.

3. Definition of order relations between intervals

3.1. Order relations for maximization problems

In this section, the order relations which represent the decision maker's preference between interval profits are defined for maximization problems. Let the uncertain profits from two alternatives be represented by intervals A and B respectively. It is assumed that the profit from each alternative is known only to lie in the corresponding interval.

First the order relation by the left and right limits of an interval is defined in Definition 3.1.

Definition 3.1. Let us define the order relation $\leq_{LR}$ between $A = [a_L, a_R]$ and $B = [b_L, b_R]$ as

$$A \leq_{LR} B \quad \text{iff} \quad a_L \leq b_L \quad \text{and} \quad a_R \leq b_R, \quad (3.1)$$

$$A <_{LR} B \quad \text{iff} \quad A \leq_{LR} B \quad \text{and} \quad A \neq B. \quad (3.2)$$

This order relation represents the decision maker's preference for the alternative with the higher minimum profit and maximum profit. It is clear from Definition 3.1 that the following properties hold for $\leq_{LR}$.

(1) If $A \leq_{LR} B$, then $a_C \leq b_C$ where a_C and b_C are the centers of A and B respectively.

(2) If $a_L = a_R$ and $b_L = b_R$, then $\leq_{LR}$ reduces to the ordinary inequality relation $\leq$ on the set of real numbers.

It is also clear that the order $\leq_{LR}$ is a partial order which is transitive, reflexive and antisymmetric. Therefore there are many pairs of intervals which can not be compared by $\leq_{LR}$. For example, if $A = [100, 200]$ and $B = [160, 180]$, then neither $A \leq_{LR} B$ nor $B \leq_{LR} A$ holds. In this case we usually prefer B . In order to represent this intuition, the order relation by the center and width of interval is defined as follows.

Definition 3.2. Let us define the order relation $\leq_{CW}$ between $A = \langle a_C, a_W \rangle$ and $B = \langle b_C, b_W \rangle$ as

$$A \leq_{CW} B \quad \text{iff} \quad a_C \leq b_C \quad \text{and} \quad a_W \geq b_W, \quad (3.3)$$

$$A <_{CW} B \quad \text{iff} \quad A \leq_{CW} B \quad \text{and} \quad A \neq B. \quad (3.4)$$

Since the center and the width of interval can be considered as the expected value and the uncer-

tainty of an interval respectively, this order relation represents the decision maker's preference for the alternative with the higher expected value and less uncertainty. This order relation is also a partial order. It is clear from Definition 3.2 that the following properties hold for $\leq_{CW}$.

(3) If $A \leq_{CW} B$, then $a_L \leq b_L$ where a_L and b_L are the left limits of A and B respectively.

(4) If $a_W = b_W = 0$, then $\leq_{CW}$ reduces to the ordinary inequality relation $\leq$ on the set of real numbers.

The order relations $\leq_{LR}$ and $\leq_{CW}$ defined above never conflict with each other in the sense that there is no such pair A and B that $A \neq B$, $A \leq_{LR} B$ and $B \leq_{CW} A$. This property is shown by the following proposition.

Proposition 3.1. If both

$$A \leq_{LR} B \quad \text{and} \quad B \leq_{CW} A \quad (3.5)$$

hold, then $A = B$.

Proof. From $A \leq_{LR} B$,

$$a_L = a_C - a_W \leq b_L = b_C - b_W \quad (3.6)$$

holds. On the other hand from $B \leq_{CW} A$,

$$b_C \leq a_C, \quad b_W \geq a_W \quad (3.7)$$

hold. Using (3.7), (3.6) can be modified as

$$a_C - a_W \leq b_C - b_W \leq a_C - b_W \leq a_C - a_W. \quad (3.8)$$

This shows that $a_C - a_W = b_C - b_W$. Since $a_L = a_C - a_W$ and $b_L = b_C - b_W$, it was shown that $a_L = b_L$. In a similar way, it can be shown that $a_R = b_R$. $\square$

3.2. Order relations for minimization problems

In this section, the order relations which represent the decision maker's preference between interval costs are defined for minimization problems. Let the uncertain costs from two alternatives be represented by intervals A and B respectively. It is assumed that the cost of each alternative is known only to lie in the corresponding interval.

First the order relation by the left and right limits of interval is defined in the same manner as Definition 3.1.

Definition 3.3. Let us define the order relation $\leq_{LR}^*$ between $A = [a_L, a_R]$ and $B = [b_L, b_R]$ as

$$A \leq_{LR}^* B \quad \text{iff } a_L \leq b_L \text{ and } a_R \leq b_R, \quad (3.9)$$

$$A <_{LR}^* B \quad \text{iff } A \leq_{LR}^* B \text{ and } A \neq B. \quad (3.10)$$

It should be noted that $\leq_{LR}^*$ by Definition 3.3 is the same order relation as $\leq_{LR}$ by Definition 3.1. The order relation $\leq_{LR}^*$ represents the decision maker's preference for the alternative with the lower minimum cost and maximum cost, that is, if $A \leq_{LR}^* B$, then A is preferred to B .

Next the order relation by the center and width of interval is defined in the following definition.

Definition 3.4. Let us define the order relation $\leq_{CW}^*$ between $A = \langle a_C, a_W \rangle$ and $B = \langle b_C, b_W \rangle$ as

$$A \leq_{CW}^* B \quad \text{iff } a_C \leq b_C \text{ and } a_W \leq b_W, \quad (3.11)$$

$$A <_{CW}^* B \quad \text{iff } A \leq_{CW}^* B \text{ and } A \neq B. \quad (3.12)$$

It should be noted that the order relation $\leq_{CW}^*$ by Definition 3.4 is different from $\leq_{CW}$ by Definition 3.2. The order relation $\leq_{CW}^*$ represents the decision maker's preference for the alternative with the lower expected cost and less uncertainty, that is: if $A \leq_{CW}^* B$, then A is preferred to B . This order relation is also a partial order. For this order relation, the following properties hold.

(5) If $A \leq_{CW}^* B$ then $a_R \leq b_R$ where a_R and b_R are the right limits of A and B respectively.

(6) If $a_W = b_W = 0$ then $\leq_{CW}^*$ reduces to the ordinary inequality relation $\leq$ on the set of real numbers.

The order relations defined in this section never conflict with each other in the sense that there is no such pair A and B that $A \neq B$, $A \leq_{LR}^* B$ and $B \leq_{CW}^* A$. This property is shown by the following proposition.

Proposition 3.2. If both

$$A \leq_{LR}^* B \quad \text{and} \quad B \leq_{CW}^* A \quad (3.13)$$

hold. Then $A = B$.

Since this proposition can be proven in a similar way as the proof of Proposition 3.1, the proof is omitted.

4. Formulation of multiobjective problems

4.1. The maximization problem

In this section, we show the reformulation of the maximization problem (1.1) as a multiobjective problem. Since the objective function $Z(x)$ of (1.1) is an interval function, it is natural that the solution set of (1.1) should be defined by preference relations between intervals. Therefore using the order relations defined in Section 3.1, which represent the decision maker's preference between interval profits, the solution set of (1.1) is defined as the nondominated solutions:

Definition 4.1. $x \in S$ is a solution of (1.1) if and only if there is no $x' \in S$ which satisfies $Z(x) <_{LR} Z(x')$ or $Z(x) <_{CW} Z(x')$.

In order to simplify this definition, first we define the order relation $\leq_{LC}$ as

$$A \leq_{LC} B \quad \text{iff } a_L \leq b_L \text{ and } a_C \leq b_C, \quad (4.1)$$

$$A <_{LC} B \quad \text{iff } A \leq_{LC} B \text{ and } A \neq B. \quad (4.2)$$

Then Proposition 4.1 holds.

Proposition 4.1. It follows that

$$A \leq_{LC} B \quad \text{iff } A \leq_{LR} B \quad \text{or} \quad A \leq_{CW} B, \quad (4.3)$$

$$A <_{LC} B \quad \text{iff } A <_{LR} B \quad \text{or} \quad A <_{CW} B. \quad (4.4)$$

Proof. Since (4.4) is easily obtained from (4.3), we only prove (4.3).

First we show that if $A \leq_{LC} B$ then $A \leq_{LR} B$ or $A \leq_{CW} B$. From $A \leq_{LC} B$,

$$a_L \leq b_L, \quad a_C \leq b_C \quad (4.5)$$

hold. If we assume that $a_W \geq b_W$, then $A \leq_{CW} B$ holds from (4.5). On the contrary, if we assume that $a_W \leq b_W$, then $A \leq_{LR} B$ holds from (4.5) and the following inequality:

$$a_R = a_L + 2a_W \leq b_L + 2b_W = b_R. \quad (4.6)$$

Therefore, it was shown that if $A \leq_{LC} B$ then $A \leq_{LR} B$ or $A \leq_{CW} B$.

Next we show that if $A \leq_{LR} B$ then $A \leq_{LC} B$. From $A \leq_{LR} B$,

$$a_L \leq b_L, \quad a_R \leq b_R \quad (4.7)$$

hold. From (4.7),

$$a_C = \frac{1}{2}(a_R + a_L) \leq \frac{1}{2}(b_R + b_L) = b_C \quad (4.8)$$

holds. Therefore $A \leq_{LC} B$ holds from (4.7) and (4.8).

Last we show that if $A \leq_{CW} B$ then $A \leq_{LC} B$. From $A \leq_{CW} B$,

$$a_C \leq b_C, \quad a_W \geq b_W \quad (4.9)$$

hold. From (4.9),

$$a_L = a_C - a_W \leq b_C - b_W = b_L \quad (4.10)$$

holds. Therefore $A \leq_{LC} B$ holds from (4.9) and (4.10). $\square$

Definition 4.1 may be simplified using this proposition as

Definition 4.2. $x \in S$ is a solution of (1.1) if and only if there is no $x' \in S$ which satisfies $Z(x) <_{LC} Z(x')$.

Let the interval coefficient A_i in (1.1) be denoted by $\langle a_{Ci}, a_{Wi} \rangle$, then the left limit $z_L(x)$ and the center $z_C(x)$ of $Z(x)$ in (1.1) may be calculated from (2.7) and (2.9) as

$$z_L(x) = (a_{C1}x_1 + \cdots + a_{Cn}x_n) - (a_{W1}|x_1| + \cdots + a_{Wn}|x_n|), \quad (4.11)$$

$$z_C(x) = a_{C1}x_1 + \cdots + a_{Cn}x_n. \quad (4.12)$$

In the case of $x \geq 0$, (4.11) may be modified as

$$z_L(x) = (a_{C1}x_1 + \cdots + a_{Cn}x_n) - (a_{W1}x_1 + \cdots + a_{Wn}x_n). \quad (4.13)$$

From the discussions so far, the solution set of (1.1) defined by Definition 4.2 can be obtained as the Pareto optimal solutions (i.e., noninferior or nondominated solutions) of the following multiobjective problem:

$$\max(z_L(x), z_C(x) : x \in S \subset R^n), \quad (4.14)$$

where $z_L(x)$ and $z_C(x)$ are calculated as (4.11) and (4.12) respectively.

4.2. The minimization problem

We can also analyze the minimization problem (1.3) in a similar way as in Section 4.1. First the solution set of (1.3) is defined in Definition 4.3.

Definition 4.3. $x \in S$ is a solution of (1.3) if and only if there is no $x' \in S$ which satisfies $Z(x') <_{LR} Z(x)$ or $Z(x') <_{CW} Z(x)$.

For the minimization problem, the order relation $\leq_{RC}^*$ is defined as

$$A \leq_{RC}^* B \quad \text{iff } a_R \leq b_R \text{ and } a_C \leq b_C, \quad (4.15)$$

$$A <_{RC}^* B \quad \text{iff } A \leq_{RC}^* B \text{ and } A \neq B. \quad (4.16)$$

Then the following proposition can be proven in a similar way as the proof of Proposition 4.1.

Proposition 4.2. It follows that

$$A \leq_{RC}^* B \quad \text{iff } A \leq_{LR}^* B \text{ or } A \leq_{CW}^* B, \quad (4.17)$$

$$A <_{RC}^* B \quad \text{iff } A <_{LR}^* B \text{ or } A <_{CW}^* B. \quad (4.18)$$

Using Proposition 4.2, Definition 4.3 may be simplified as follows:

Definition 4.4. $x \in S$ is a solution of (1.3) if and only if there is no $x' \in S$ which satisfies $Z(x') <_{RC}^* Z(x)$.

The right limit $z_R(x)$ of the interval objective function $Z(x)$ in (1.3) may be calculated from (2.7) and (2.9) as

$$z_R(x) = (a_{C1}x_1 + \cdots + a_{Cn}x_n) + (a_{W1}|x_1| + \cdots + a_{Wn}|x_n|), \quad (4.19)$$

where a_{Ci} is the center and a_{Wi} is the width of the coefficient A_i of $Z(x)$. In case of $x \geq 0$, (4.19) can be modified as

$$z_R(x) = (a_{C1}x_1 + \cdots + a_{Cn}x_n) + (a_{W1}x_1 + \cdots + a_{Wn}x_n). \quad (4.20)$$

The solution set of (1.3) defined by Definition 4.4 can be obtained as the Pareto optimal solutions of the following multiobjective problem:

$$\min((z_R(x), z_C(x)) : x \in S \subset R^n), \quad (4.21)$$

where $z_R(x)$ and $z_C(x)$ are calculated as (4.19) and (4.12) respectively. It should be noted that the objective functions of (4.21) are to minimize $z_R(x)$ and $z_C(x)$ whereas those of (4.14) are to maximize $z_L(x)$ and $z_C(x)$.

Lastly, let us discuss the connection between the maximization problem (1.1) and the minimization problem (1.3). In the ordinary mathematical programming problem with the single valued objective function $z(x)$, maximizing $z(x)$ is equivalent to minimizing $-z(x)$. It is shown by the

following proposition that the same property holds for our problems.

Proposition 4.3. *The following two problems with interval objective functions have the same solution set in the sense of Definition 4.2 and 4.4.*

$$\max(Z(x) = A_1x_1 + A_2x_2 + \cdots + A_nx_n : x \in S \subset R^n), \quad (1.1)$$

$$\min(-Z(x) = -A_1x_1 - A_2x_2 - \cdots - A_nx_n : x \in S \subset R^n). \quad (4.22)$$

Proof. This proposition can be proven by showing the following equivalent relation:

$$Z(x) <_{LC} Z(x') \quad \text{iff} \quad -Z(x') <_{RC}^* -Z(x). \quad (4.23)$$

In order to prove (4.23), let us prove the following equivalent relation:

$$Z(x) <_{LC} Z(x') \quad \text{iff} \quad Y(x') <_{RC}^* Y(x), \quad (4.24)$$

where $Y(x) = -Z(x)$ and $Y(x') = -Z(x')$. This relation is clear from the definitions of $<_{LC}$ and $<_{RC}^*$ and the following relations:

$$y_R(x) = -z_L(x), \quad y_R(x') = -z_L(x'), \quad (4.25)$$

$$y_C(x) = -z_C(x), \quad y_C(x') = -z_C(x'). \quad \square \quad (4.26)$$

5. Numerical examples

In this section two numerical examples are shown to illustrate our method proposed for dealing with the interval objective function. First let us consider the product-mix problem where the profit from each product to be selected is uncertain.

Example 1. Let us assume that the uncertain profit from the i -th product is estimated by an interval A_i ($i = 1, 2, \dots, n$). It is also assumed that the production time required for the i -th product is b_i and the total available production time is s . Then our problem is to maximize the total profit by selecting the products to be produced in the available production time s :

$$\max. \quad Z(x) = A_1x_1 + A_2x_2 + \cdots + A_nx_n \quad (5.1)$$

subject to

$$b_1x_1 + b_2x_2 + \cdots + b_nx_n \leq s, \quad (5.2)$$

$$x_i = 0 \text{ or } 1, \quad i = 1, 2, \dots, n. \quad (5.3)$$

Let us consider the following numerical example.

[Problem 1]

$$\begin{aligned} \max. \quad Z(x) = & [1000, 1140]x_1 + [750, 820]x_2 \\ & + [700, 850]x_3 + [600, 675]x_4 \\ & + [500, 720]x_5 + [345, 1020]x_6 \end{aligned} \quad (5.4)$$

subject to

$$\begin{aligned} 110x_1 + 80x_2 + 88x_3 + 68x_4 + 72x_5 + 72x_6 \\ \leq 230, \end{aligned} \quad (5.5)$$

$$x_i = 0 \text{ or } 1, \quad i = 1, 2, 3, 4, 5, 6. \quad (5.6)$$

The solution set of this problem is obtained from the following multiobjective problem:

[MP1]

$$\max. \quad (z_L(x), z_C(x)) \quad (5.7)$$

subject to (5.5) and (5.6),

where

$$\begin{aligned} z_L(x) = & 1000x_1 + 750x_2 + 700x_3 + 600x_4 \\ & + 500x_5 + 345x_6, \end{aligned} \quad (5.8)$$

$$\begin{aligned} z_C(x) = & 1070x_1 + 785x_2 + 775x_3 + 637.5x_4 \\ & + 610x_5 + 682.5x_6. \end{aligned} \quad (5.9)$$

Since the size of this problem is very small, all of the Pareto optimal solutions are calculated by the enumeration method as

$$\{x^a = (0, 1, 0, 1, 1, 0), x^b = (0, 1, 0, 1, 0, 1)\}.$$

The corresponding intervals are calculated as

$$Z(x^a) = [1850, 2215] = \langle 2032.5, 182.5 \rangle, \quad (5.10)$$

$$Z(x^b) = [1695, 2515] = \langle 2105, 410 \rangle. \quad (5.11)$$

If we select the products 2, 4 and 5, that is, x^a , then the minimum profit is high but the expected value is low. On the other hand, if we select the products 2, 4 and 6, that is, x^b , then the expected value is high but the minimum profit is low. Since these two intervals can not be compared by the order relations defined in Section 3.1, the selection from the two alternatives x^a and x^b is entirely depending on the decision maker.

Example 2. Let us consider the following linear programming problem with the interval objective function:

[Problem 2]

$$\begin{aligned} \max. \quad Z(x) = [15, 17]x_1 + [15, 20]x_2 \\ + [10, 30]x_3 \end{aligned} \quad (5.12)$$

subject to

$$4.6x_1 + 7.6x_2 + 3.6x_3 \leq 21, \quad (5.13)$$

$$5.8x_1 + 3.6x_2 + 7.8x_3 \leq 31, \quad (5.14)$$

$$7.5x_1 + 6.5x_2 + 6.8x_3 \leq 41, \quad (5.15)$$

$$x_1, x_2, x_3 \geq 0. \quad (5.16)$$

The solution set of this problem is the Pareto optimal solutions of the following multiobjective problem:

[MP2]

$$\max. \quad (z_L(x), z_C(x)) \quad (5.17)$$

subject to (5.13), (5.14), (5.15) and (5.16),

where

$$z_L(x) = 15x_1 + 15x_2 + 10x_3, \quad (5.18)$$

$$z_C(x) = 16x_1 + 17.5x_2 + 20x_3. \quad (5.19)$$

Using the weighting method [3], we can obtain the Pareto optimal solutions of MP2 from the following problem:

[LP2]

$$\max. \quad wz_L(x) + (1 - w)z_C(x) \quad (5.20)$$

subject to (5.13), (5.14), (5.15) and (5.16).

The Pareto optimal solutions of MP2 are obtained from LP2 by gradually increasing w in (5.20) from 0 to 1 as follows:

$$\begin{aligned} \{x^a = (0, 1.13, 3.45), x^b = (3.48, 0, 1.39), \\ x^c = (4.57, 0, 0)\}. \end{aligned}$$

The corresponding intervals are calculated as

$$Z(x^a) = [51.4, 126.2] = \langle 88.8, 37.4 \rangle, \quad (5.21)$$

$$Z(x^b) = [66.1, 100.8] = \langle 83.4, 17.3 \rangle, \quad (5.22)$$

$$Z(x^c) = [68.5, 77.6] = \langle 73.0, 4.57 \rangle. \quad (5.23)$$

6. Conclusion

In this paper the maximization problem with the interval objective function was converted into the multiobjective problem with two objectives which are to maximize the left limit and the center of the interval objective function. These two objectives can be considered as the maximization of the worst case and the average case respectively. The minimization problem with the interval objective function was also converted into the multiobjective problem whose objectives are to minimize the right limit and the center of the interval objective function. These two objective functions can be considered as the minimization of the worst case and the average case. The solution sets of our optimization problems with the interval objective functions are defined as the Pareto optimal solutions of the corresponding multiobjective problems. Therefore the solution set defined in this paper includes the optimal solutions against the worst case and against the average case. The proposed method for dealing with the interval objective function was illustrated by the numerical examples.

References

- [1] Alefeld, G., and Herzberger, J., *Introduction to Interval Computations* (Translated by J. Rokne), Academic Press, New York, 1983.
- [2] Bellman, R.E., and Zadeh, L.A., "Decision-making in a fuzzy environment", *Management Science* 17 (1970) 141–164.
- [3] Chankong, V., and Haimes, Y.Y., *Multiobjective Decision Making: Theory and Methodology*, North-Holland, New York, 1983.
- [4] Charnes, A., and Cooper, W.W., "Chance constrained programming", *Management Science* 6 (1959) 73–79.
- [5] Falk, J.E., "Exact solutions of inexact linear programs", *Operations Research* 24 (1976) 783–787.
- [6] Moore, R.E., *Method and Applications of Interval Analysis*, SIAM, Philadelphia, 1979.
- [7] Sengupta, J.K., *Stochastic Programming: Methods and Applications*, North-Holland, Amsterdam, 1972.
- [8] Soyster, A.L., "Inexact linear programming with generalized resource sets", *European Journal of Operational Research* 3 (1979) 316–321.
- [9] Zimmerman, H.-J., "Description and optimization of fuzzy systems", *International Journal of General Systems* 2 (1976) 209–215.